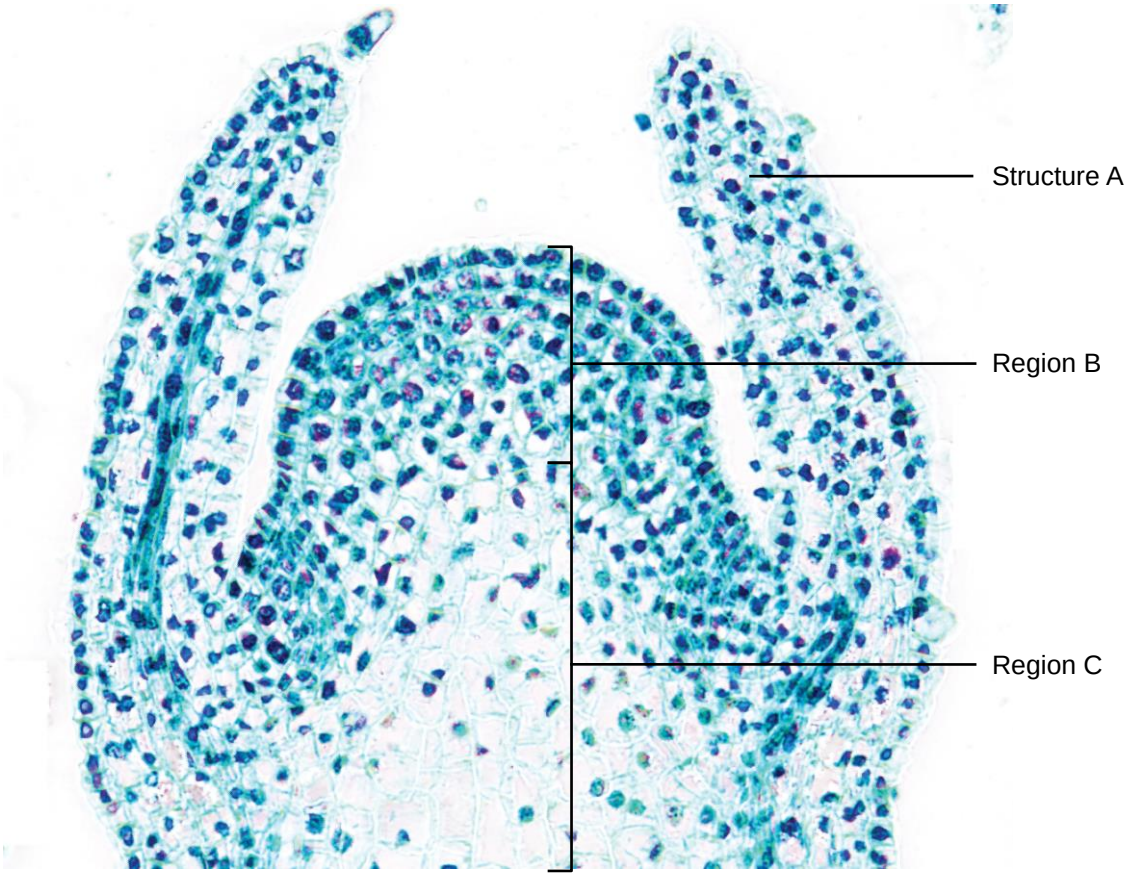
Structured questions

|!|BCELB0150000001|!| ★

The photomicrograph below shows the longitudinal section of the shoot tip of a plant.



Shoot tip, L.S. (×100)

(a) Describe and explain how the growth of the youth seedling would be affected if structure A is detached from the shoot tip. (3 marks)

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(b) (i) With reference to the photomicrograph, suggest and explain which region (region B or C) belongs to the region of elongation. (2 marks)

(ii) Describe how the cells in the region of elongation help increase the length of the shoot. (2 marks)

##

(a) Structure A will develop into young leaves. (1)
If structure A is detached from the shoot, the young seedling cannot carry out photosynthesis to produce food for growth. (1)
Therefore, the young seedling may have stunted growth. (1)

(b) (i) Region C (1)
The cells in region C are larger than that of region B. (1)

(ii) Cells in region C absorb water by osmosis to form many small vacuoles in the cytoplasm. (1)
This stretches the thin cell walls and the cells elongate, hence increasing the length of the shoot. (1)

(7 marks)

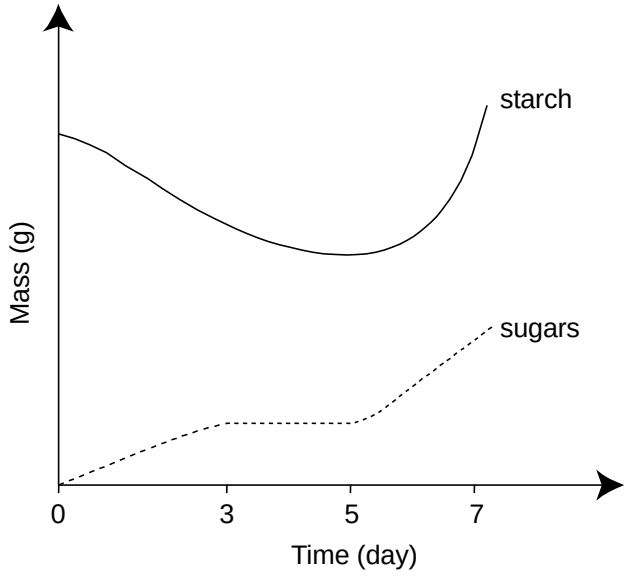
##

	Skill assessed	Marks
(a)	Applying concepts	3
(b)(i)	Analysing data	2
(b)(ii)	Applying concepts	2

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|!|BCELB0150000002|!| ★

The following graph shows the changes in the content of starch and sugars of a germinating seed during a week.



(a) Explain why the seed contains no sugar on day 0. (1 mark)

(b) Explain the changes in the mass of starch in the seed during the week. (5 marks)

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(c) The growth of the seed can be determined by measuring its fresh mass.

(i) Draw a graph to show the changes in the fresh mass of the seed during the week in the space below. (3 marks)

(ii) State ***one*** disadvantage for using fresh mass as an indicator of growth. (1 mark)

##

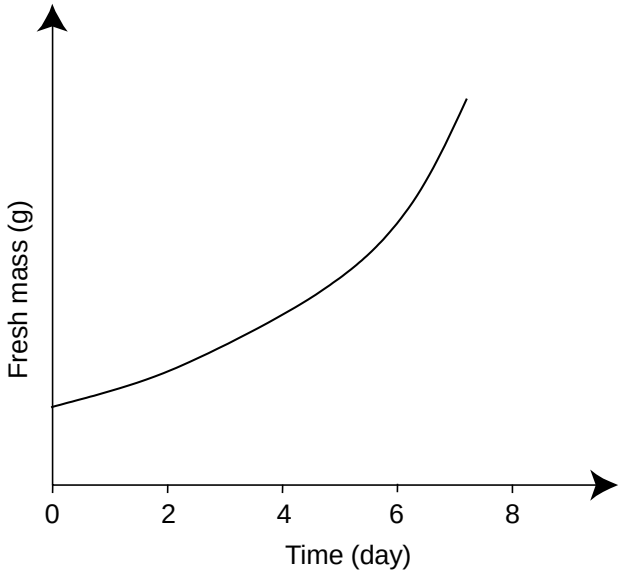
(a) This is because the cotyledons of the seed contain food reserves that are composed mainly of starch and contain no sugars. (1)

(b) The starch level decreases from day 0 to day 5 (1)
as the starch stored in the cotyledon is broken down into sugars for respiration to provide energy. (1)
Dry mass of starch increases from day 5 onwards (1)
because sugar is synthesized in the leaf by photosynthesis. (1)
The sugar is then converted to starch for storage. (1)

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(c) (i)

Changes in the fresh mass of the seed during the week



Correct curve (a continuous curve without decrease in weight) (1)

Correct choice of labelled axes (1)

Correct title (1)

(ii) The fresh mass varies with water content which is affected by environmental conditions / the behaviour of the organism. (1)

(10 marks)

##

	Skill assessed	Marks
(a)	Applying concepts	1
(b)	Analysing data	2
	Applying concepts	3
(c)(i)	Presenting information	3
(c)(ii)	Understanding basic concepts	1

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|!|BCELC0150000003|!|

Angela planted 20 mung beans in the school garden. Only 15 germinated to become seedlings.

(a) Besides water and oxygen, state another factor that is necessary for seed germination. (1 mark)

(b) Calculate the percentage of seed germination. Show your working. (2 marks)

(c) Describe how the seed uses the stored starch in germination. (3 marks)

(d) The total number of cells in the seedling increases with time.

(i) State the type of cell division involved. (1 mark)

(ii) State ***two*** plant tissues where the cell division mentioned in (i) takes place. (2 marks)

##

(a) Optimum temperature / warmth (Or other reasonable answers) (1)

(b) Percentage of seed germination = (15/20) x 100% (1)

= 75% (1)

(c) The starch has to be hydrolysed by enzyme into sugars. (1)

The sugars are transported to the plumule and the radicle (1)

where oxidation / respiration takes place to release energy for growth / synthesis of new cell wall. (1)

(d) (i) Mitotic cell division (1)

(ii) Apical meristem (1)

Lateral meristem / vascular cambium (1)

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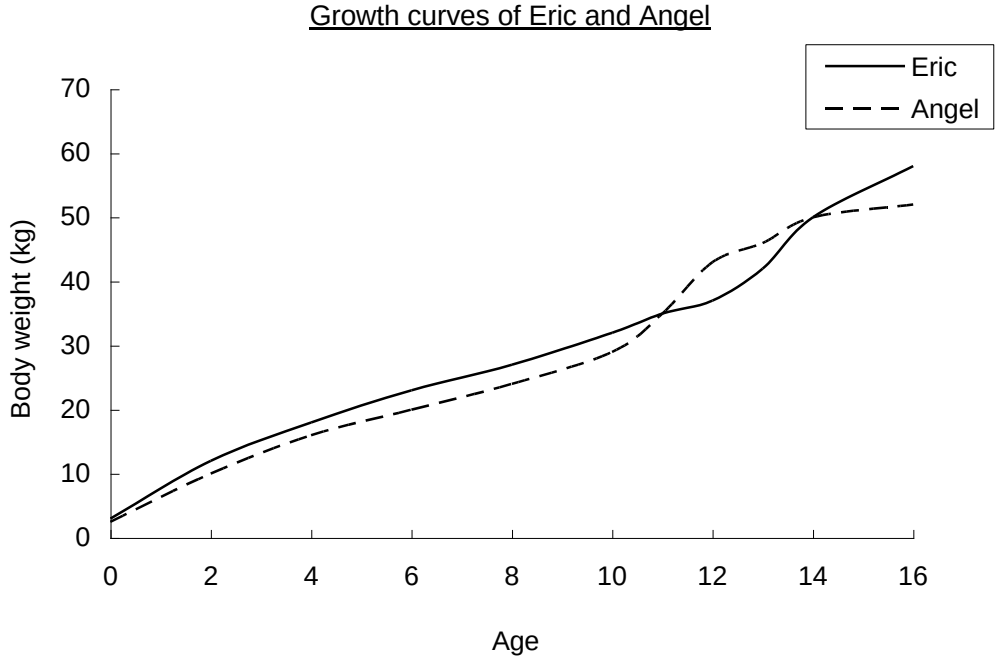
(9 marks)

##

	Skill assessed	Marks
(a)	Understanding basic concepts	1
(b)	Mathematical skills	2
(c)	Understanding basic concepts	3
(d)(i)	Understanding basic concepts	1
(d)(ii)	Understanding basic concepts	2

|!|BCELB0150000004|!| ★ **Cross topic**

Two fraternal twins, Eric and Angel, are 16 years old. Their mother kept the record of the changes of their body weights. The following graph shows their body weights over the years.



(a) Describe the occurrence of the fraternal twins. (2 marks)

(b) According to the graph, describe the difference in the growth rates of the twins:

(i) between the ages of 11 and 14; (1 mark)

(ii) after the age of 14. (1 mark)

(c) The growth and development of humans are regulated by hormones, secreted by endocrine gland.

(i) Which part of the body has the highest growth rate at age 2? Name the endocrine gland involved. (2 marks)

(ii) Which part of the body has the highest growth rate at age 13? Name the endocrine gland involved. (2 marks)

##

(a) When two ova are released from the ovary at the same time (1)
and are fertilized independently by two sperms. (1)
Fraternal twins will be resulted.

(b) (i) Angel grew faster than Eric. (1)

(ii) Eric grew faster than Angel. (1)

(c) (i) Brain (or other reasonable answers) (1)
Endocrine gland: pituitary gland (which secretes the growth hormone) (1)

(ii) Reproductive organs (or other reasonable answers) (1)
Endocrine gland: testes (for male) and ovary (for female) (1)

(8 marks)

##

	Skill assessed	Marks
(a)	Understanding basic concepts	2
(b)(i)	Analysing data	1
(b)(ii)	Analysing data	1
(c)(i)	Understanding basic concepts	2
(c)(ii)	Understanding basic concepts	2

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|!|BCELB0150000005|!| ★ **SI**

Tina and her classmates carried out an investigation on castor oil seeds. They allowed the seeds to germinate and then analysed the lipid and glucose contents in the seeds.

The following table shows the changes in the lipid and glucose contents of castor oil seedlings during 17 days.

Time of germination (day)	**Lipid content (% of dry mass)**	**Glucose content (% of dry mass)**
0	70.9	0.0
2	63.5	0.0
5	47.8	3.0
8	24.8	7.9
11	17.7	8.8
14	13.0	13.1
17	4.8	17.9

(a) Based on the results in the table, state the main form of stored organic food in castor oil seed. (1 mark)

(b) Explain the changes of lipid content in the seedlings. (4 marks)

(c) Explain the changes of glucose content in the seedlings. (3 marks)

(d) Dry mass was measured in the project. State ***two*** disadvantages of measuring the growth by measuring the dry mass. (2 marks)

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##

(a) Lipid (1)

(b) During germination, lipid is broken down into fatty acids and glycerol (1)
which are oxidized in respiration to release energy. (1)
The fatty acids, glycerol and the energy are used in forming new cells at the growing regions. (1)
This leads to a drop in lipid content of the seedlings. (1)

(c) Green leaves are not yet formed by the seedlings until day 5. (1)
When the leaves are grown, the seedlings carry out photosynthesis to make glucose. (1)
So, the glucose content of the seedlings starts to increase on day 5. (1)

(d) Any ***two*** of the following: (1 mark each) (2)
- The seeds must be killed for analysis.
- A large number of seeds are required in the measurement.
- It is time-consuming.
(Or other reasonable answers)

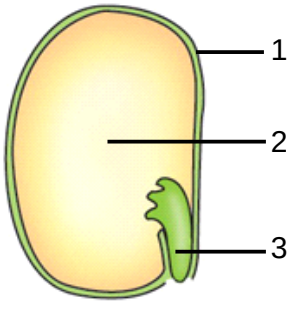
(10 marks)

##

	Skill assessed	Marks
(a)	Analysing data	1
(b)	Applying concepts	4
(c)	Applying concepts	3
(d)	Understanding basic concepts	2

|!|BCELC0150000006|!|

The following diagram shows the structure of a seed.



(a) State ***one*** function of structure 1. (1 mark)

(b) Name structure 2. (1 mark)

(c) Amylase and protease activities are detected when the seed germinates. Explain the significance of these enzyme activities to the germination of the seed.(5 marks)

(d) Structure 3 will develop into a root. State and explain ***one*** structure of the root that can help increase water absorption from the soil. (2 marks)

##

(a) To protect the embryo from (mechanical) damage /
To protect the embryo from attack by microorganisms (1)

(b) Cotyledon (1)

(c) Amylase breaks down the stored starch into maltose which is converted to glucose. (1)
Glucose is used in respiration to release energy. (1)
Glucose is also converted to cellulose for cell wall synthesis. (1)
Proteases break down the stored proteins into amino acids (1)
which can be used to synthesize other proteins such as enzymes / proteins in the cell membrane and cytoplasm. (1)

(d) Root hairs (1)
Root hairs increase the surface area for water absorption. (1)

(9 marks)

##

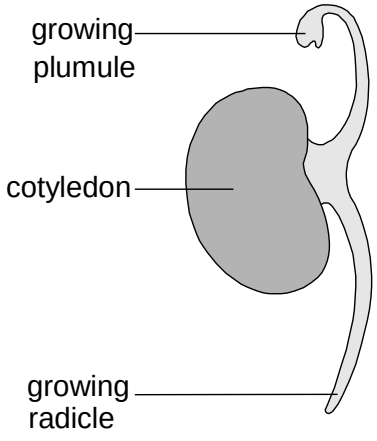
	Skill assessed	Marks
(a)	Understanding basic concepts	1
(b)	Analysing data	1
(c)	Understanding basic concepts	5

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(d)	Understanding basic concepts	2

|!|BCELB0150000007|!| ★ **Cross topic**

The diagram below shows the structure of a germinating seed.



(a) State ***two*** reasons to explain why water is essential for the germination of a seed. (2 marks)

(b) Lipid stored in the seed is broken down by lipase for growth. State how lipid in the cotyledon can be used in the growth of the plumule. (3 marks)

(c) What is the tropic response of the radicle to gravity? Explain a significance of such response to the plant. (2 marks)

##

(a) Any ***two*** of the following: (1 mark each) (2)

- to soften the seed coat for the protrusion of the radicle

- to make the cotyledon swell for breaking the seed coat
- to activate enzymes for metabolism
- to hydrolyse food substances into simple soluble form for growth
- to act as a transport medium of food substances to the growing plumule and radicle

(b) Lipid is hydrolysed by lipase to form fatty acids and glycerol (1)
which are transported to the growing point of the plumule. (1)
The fatty acids and glycerol are used to synthesize the cell membrane (or other correct examples) of new cells. (1)

(c) The radicle grows downwards. (1)
This enables the radicle to grow deeper into the soil for better anchorage / to obtain an adequate amount of water and dissolved minerals from the soil. (1)

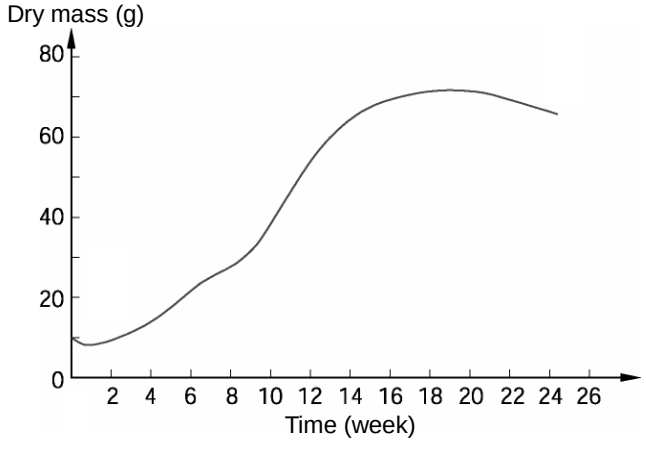
(7 marks)

##

	Skill assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	3
(c)	Understanding basic concepts	2

|!|BCELB0150000008|!| ★

The following graph shows the changes in dry mass of an annual plant after the onset of germination.



(a) Account for the initial decrease in dry mass of the plant. (2 marks)

(b) Explain the changes in the dry mass of the plant from week 2 to week 18. (4 marks)

(c) From around week 21 onwards, there was a drop in the dry mass of the plant Give ***one*** reason to explain this phenomenon. (1 mark)

(d) In week 24, the plant was dead and degraded by some organisms. State ***one*** group of organisms involved in the process of degradation. (1 mark)

##

(a) It was because the food reserve was consumed by respiration (1)
to release energy for growth. (1)

(b) From week 2 to week 18, the plant developed green leaves (1)
The leaves carried out photosynthesis to make food. (1)
When the rate of photosynthesis was higher than that of respiration / the rate of food production was higher than that of food consumption, excess food was stored. (1)
Therefore, the dry mass increased from week 2 to week 18. (1)

(c) Dispersal of fruits / seeds (1)

(d) Fungi / decomposers (Or other reasonable answers) (1)

(8 marks)

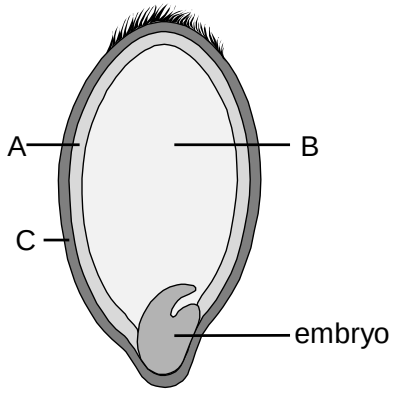
##

	Skill assessed	Marks
(a)	Applying concepts	2
(b)	Analysing data	1
	Applying concepts	3
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1

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|!|BCELC0150000009|!|

The diagram below shows the structure of a germinating barley grain.



(a) After water is absorbed from the soil into the barley grain, the embryo begins to synthesize a plant hormone that can activate genes in the cells of structure A for the synthesis of enzymes.

(i) Describe how this plant hormone allows the barley grain to utilize the starch stored in structure B to form maltose. (4 marks)

(ii) Describe how maltose can be utilized for growth in embryo. (4 marks)

(b) Structure C is formed from the fusion of seed coat and fruit wall. Which flower parts are the seed coat and fruit wall formed from? (2 marks)

##

(a) (i) The plant hormone produced by the embryo diffuses to structure A. (1)
Amylase synthesis is activated, resulting in the production of amylase. (1)
Amylase then diffuses to structure B. (1)

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Amylase hydrolyses starch in structure B into maltose. (1)

(ii) Maltose is hydrolysed by maltase to glucose (1)

which diffuses to the embryo. (1)

Glucose is then utilized for releasing energy by respiration (1)

and as a raw material for cell wall formation (or other reasonable examples). (1)

(b) Seed coat: integument (1)

Fruit wall: the wall of ovary (1)

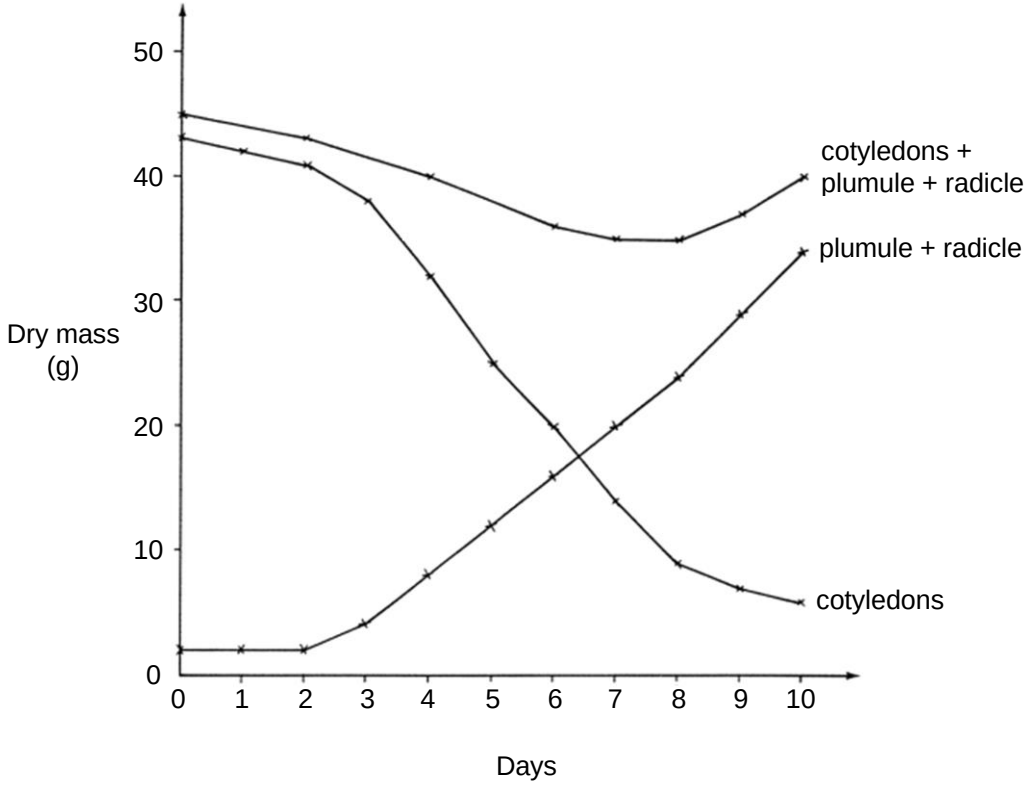
(10 marks)

##

	Skill assessed	Marks
(a)(i)	Understanding basic concepts	4
(a)(ii)	Understanding basic concepts	4
(b)	Understanding basic concepts	2

|!|BCELB0150000010|!| ★

In order to study the germination and early growth of mung bean seeds, a number of mung beans were sown in soil. The dry mass of different parts was measured at regular intervals. The results are shown in the following graph.



(a) State ***two*** functions of the cotyledon in the seed. (2 marks)

(b) Explain why there is a decrease in the dry mass of the cotyledons during the germination of seeds. (2 marks)

(c) Account for the curve of the total dry mass (cotyledons + plumule + radicle). (6 marks)

##

(a) It contains the food reserves. (1)
It contains enzymes to digest the stored food into simpler molecules that are used for growth. (1)

(b) During the germination of seeds, the food substances stored in the cotyledons are broken down to release energy (1)
to support the growth of the plumule and radicle. (1)

(c) During the first seven days, the stored food substance is broken down in respiration to release energy to support growth. (1)
Once the green aerial part has appeared, (1)
photosynthesis takes place and carbohydrates are produced. (1)
When the rate of photosynthesis is higher than the rate of respiration, the production of food is faster than the consumption of food. (1)
This results in a net gain in the amount of organic materials. (1)
As a result, there is an increase in the total dry mass. (1)

(10 marks)

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##

	Skill assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	2
(c)	Applying concepts	6

|!|BCELB0150000011|!| ★

The following table shows the height of a male that was measured at two-year-intervals from birth to 22 years old.

Age (years)	Height (cm)
Birth	52
2	80
4	96
6	110
8	118
10	128
12	140
14	160
16	180
18	186
20	189
22	189

(a) Plot a graph using the above data in the space below. (3 marks)

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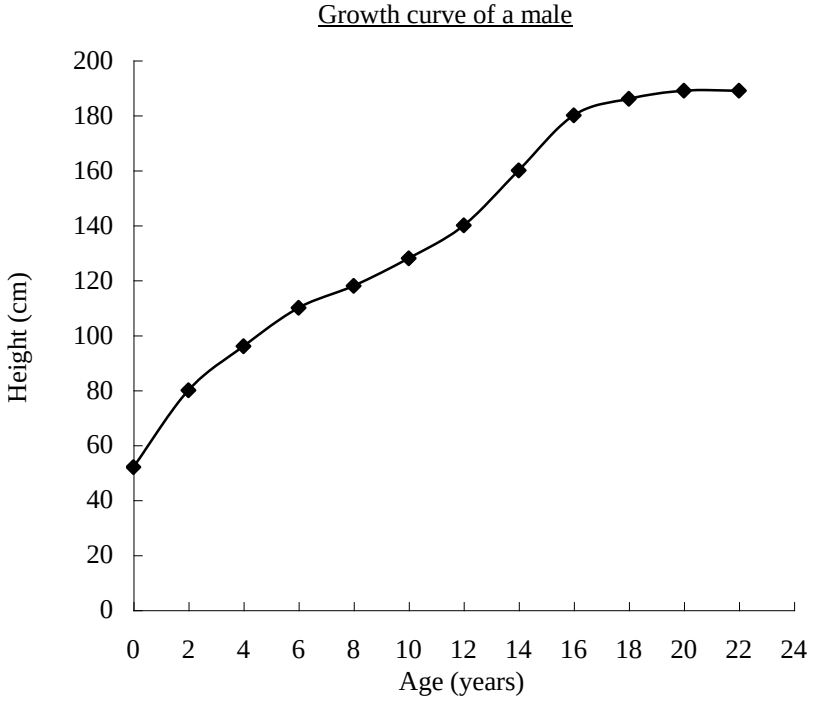
(b) From the graph, find out in which period(s) the increase in height was the fastest in the male. (2 marks)

(c) (i) State ***two*** parts of the male body that grow rapidly from age 12 to 16. (2 marks)

(ii) Name ***one*** hormone that contributes to the growth you answered in (i). In which part of the body is this hormone produced? (2 marks)

##

(a)



Correct choice of labelled axes (1)
Correct plotting and joining of points (1)
Correct title (1)

(b) From birth to two years old (1)
From 12 to 14 years old (1)

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(c) (i) The reproductive organ (1)

The arms and legs (1)

(Or other reasonable answers)

(ii) Any ***one*** set of the following: (2)

- Male sex hormone: produced in testes
- Growth hormone: produced in pituitary gland

(Or other reasonable answers)

(9 marks)

##

	Skill assessed	Marks
(a)	Analysing data	3
(b)	Analysing data	2
(c)(i)	Understanding basic concepts	2
(c)(ii)	Understanding basic concepts	2

|!|BCELB0150000012|!| ★

In an investigation, certain bean seeds were allowed to germinate and develop into young plants. The dry mass was measured at 5-day intervals over a period of 40 days. The results are shown in the table below.

Time (day)	Dry mass (g)
0	4.8
5	4.3
10	3.5
15	3.1
20	3.0
25	3.1
30	3.3
35	3.9
40	5.5

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(a) Plot a graph in the space below to show the changes in the dry mass of the plant with time. (3 marks)

(b) Explain the changes in the dry mass between day 1 and day 20 in the experiment. (2 marks)

(c) (i) According to the graph, from which day did the dry mass of the plants increase? (1 mark)

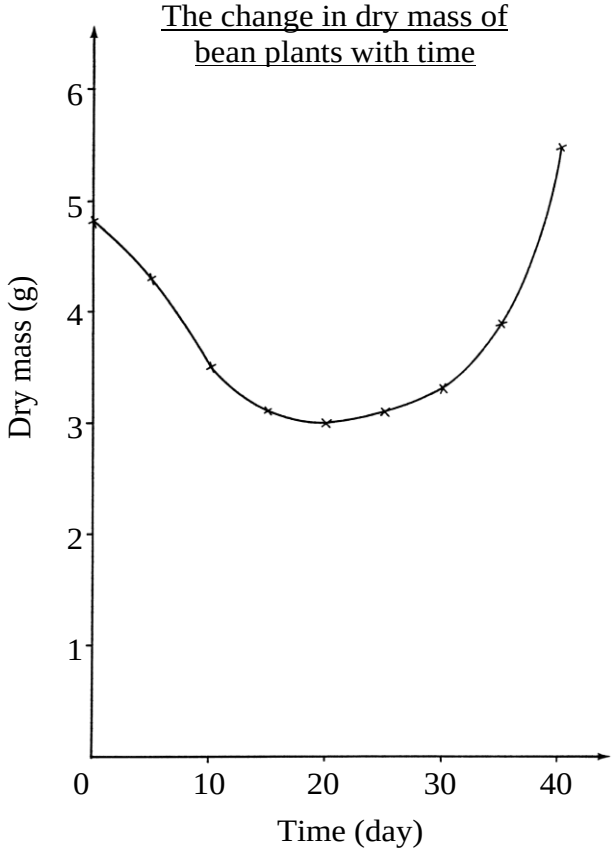
(ii) Explain the increase in the dry mass of the plants. (3 marks)

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(d) If fresh mass were measured instead of dry mass in this experiment, would the results be different? Explain. (2 marks)

##

(a)



Correct choice of labelled axes (1)
Correct plotting and joining of points (1)
Correct title (1)

(b) The stored food substances in the seeds were broken down by respiration to release energy to support growth. (1)
Therefore the dry mass of the seeds / plants decreased. (1)

(c) (i) Day 20 (1)
(ii) The plant developed green leaves. (1)
This enabled the plant to carry out photosynthesis. (1)
The photosynthetic rate gradually exceeded the rate of respiration. (1)
Therefore, the amount of food substance produced by photosynthesis was larger than that consumed by respiration, and the dry mass increased.

(d) The fresh mass of the plants would increase continuously throughout the experiment (1)

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because water was continuously absorbed for the growth of the plants. (1)

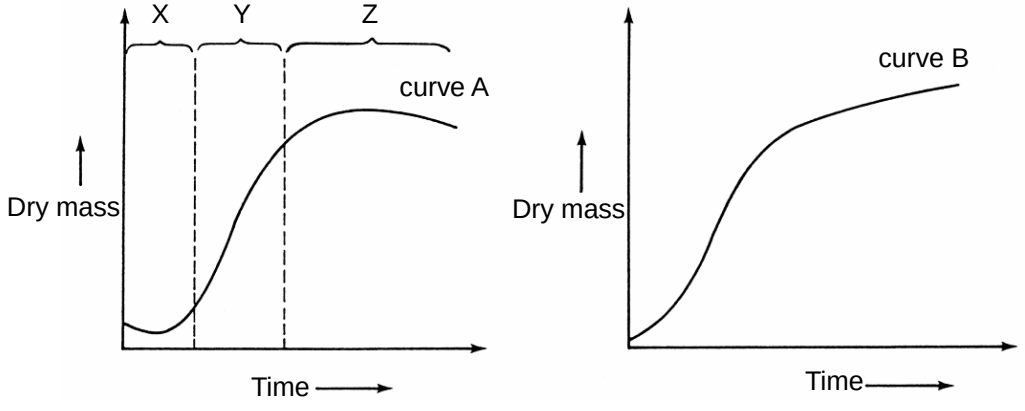
(11 marks)

##

	Skill assessed	Marks
(a)	Presenting information	3
(b)	Analysing data	1
	Applying concepts	1
(c)(i)	Analysing data	1
(c)(ii)	Applying concepts	3
(d)	Applying concepts	2

|!|BCELB0150000013|!| ★

The following graphs show the growth curves (curves A and B) of two living organisms.



(a) Which curve do you think is the growth curve of a plant? Explain your answer. (3 marks)

(b) List ***two*** environmental factors that affect plant growth. (2 marks)

(c) State ***two*** differences between growth curves A and B. (2 marks)

(d) State ***two*** differences between plant growth and human growth. (2 marks)

##

(a) Curve A (1)

During seed germination, the stored food substance in the seed is broken down to release energy to support the growth of the seed. (1)

Therefore, there is a decrease in the weight of the organism at the beginning. (1)

(b) Any ***two*** of the following: (1 mark each) (2)

- Light
- Temperature
- Water availability
- Mineral content in soil

(Or other reasonable answers)

(c)

Growth curve A	**Growth curve B**	
There is a decrease in the dry mass at the beginning.	There is no decrease in the dry mass at the beginning.	(1)
There is a decline in the dry mass at the end.	There is a steady increase in the dry mass at the end.	(1)

(d) Any ***two*** of the following: (1 mark each) (2)

	Plant growth	**Human growth**
Period of growth	Growth continues throughout life.	There is no net growth when maturity is reached.
Region of growth	in meristems	in all parts of the body
Growth spurts	one	two

(Or other reasonable answers)

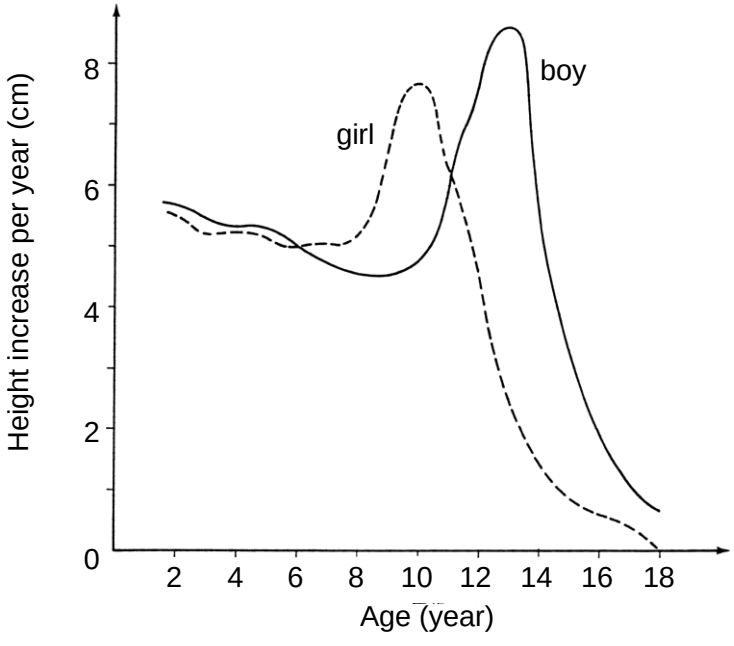
(9 marks)

##

	Skill assessed	Marks
(a)	Analysing data	1
	Applying concepts	2
(b)	Understanding basic concepts	2
(c)	Analysing data	2
(d)	Applying concepts	2

|!|BCELB0150000014|!| ★ **Cross topic**

The following graph shows the height increase of a boy and a girl from age 1 to age 18. The data were recorded in each year.



(a) At which age(s) do boys and girls have the same height increase? (1 mark)

(b) According to the graph, state ***two*** differences in growth between the boy and the girl. (2 marks)

(c) (i) During which stage of growth is the height increase the greatest in both boys and girls? (1 mark)

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(ii) Besides the increase in height, describe ***two*** other physical changes in the boy during this stage. (2 marks)

(iii) Name ***two*** food substances that need to be taken in larger amounts in order to support the growth of the girl during the stage answered in (i). Explain why. (4 marks)

##

(a) Age 6 and age 11 (1)

(b) Any ***two*** of the following comparisons: (1 mark each) (2)

Growth of the boy	Growth of the girl
- higher maximal height increase per year	- lower maximal height increase per year
- rapid growth begins later	- rapid growth begins earlier

(Or other reasonable answers)

(c) (i) Puberty (1)

(ii) Any ***two*** of the following: (1 mark each) (2)

- Growth of facial hair and body hair, including underarm, abdominal, chest hair and pubic hair.
- Enlargement of larynx (Adam's apple) and deepening of voice
- Broadening of shoulders and chest, and increase in muscle mass
- Enlargement of the penis
- Increased secretions of oil and sweat glands, often causing acne and body odour

(Or other reasonable answers)

(iii) Any ***two*** sets of the following: (2 marks each) (4)

Food substance	Reason
1. Protein	for the synthesis of new cells because it constitutes the cell membrane and cytoplasm

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2. Calcium	for the formation of bones, this leads to lengthening of limbs
3. Iron	for the formation of haemoglobin and hence red blood cells and this can replenish blood is lost during menstruation

(Or other reasonable answers)

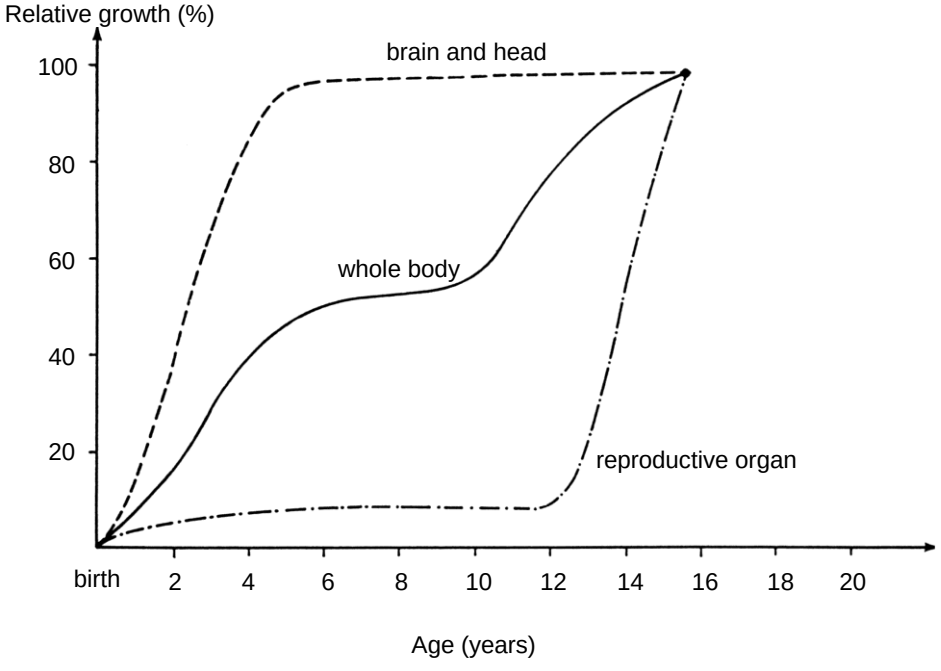
(10 marks)

##

	Skill assessed	Marks
(a)	Analysing data	1
(b)	Analysing data	2
(c)(i)	Analysing data	1
(c)(ii)	Understanding basic concepts	2
(c)(iii)	Understanding basic concepts	4

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The following graph illustrates the growth of different parts of the body of a girl at different ages.



(a) In which age range did the brain and head grow the fastest? (1 mark)

(b) (i) When did puberty start? (1 mark)

(ii) State ***two*** physiological changes that took place when the girl was at the stage of puberty. (2 marks)

(iii) What causes the rapid growth of reproductive organs in the girl? (2 marks)

(c) What is the advantage of having late development of the reproductive organs in the human body? (1 mark)

##

(a) From birth to age 5 (1)

(b) (i) At age 12 (1)

(ii) Any ***two*** of the following: (1 mark each) (2)

- growth and development of breasts
- widening of hips / pelvic girdle
- more fat deposits under skin
- growth of pubic hair

(Or other reasonable answers)

(iii) The ovaries secrete the female sex hormones (1)

which stimulate the development of sexual characteristics in females. (1)

(c) They can take care of their children after they become physically and mentally mature. (1)

(7 marks)

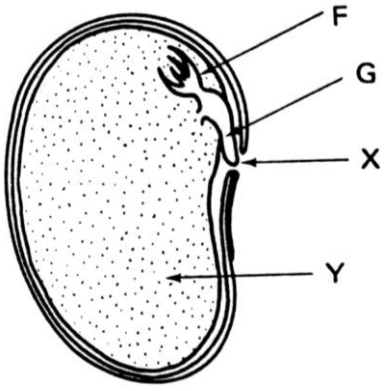
##

	Skill assessed	Marks
(a)	Analysing data	1
(b)(i)	Understanding basic concepts	1
(b)(ii)	Understanding basic concepts	2
(b)(iii)	Understanding basic concepts	1
(c)	Understanding basic concepts	1

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|!|BCELB0150000016|!| ★ **Cross topic**

The diagram below shows the longitudinal section of a pea.



(a) Name structure X and state its function. (2 marks)

(b) Name structure Y and state its function. (2 marks)

(c) State ***one*** function of water and oxygen in the germination of seeds. (2 marks)

(d) Why is seed dispersal necessary? (2 marks)

(e) Describe the method of seed dispersal of peas. (2 marks)

(f) Which structure, F or G, can show positive phototropism? (1 mark)

##

(a) X: micropyle (1)

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It allows the entry of water and oxygen into the seed. (1)

(b) Y: cotyledon (1)

It contains food reserves for the growth of the plumule and radicle. (1)

(c) Any ***one*** function of water: (1)

- activates the enzymes
- as a medium for chemical and enzyme reactions
- softens the seed coat

(Or other reasonable answers)

Function of oxygen: to oxidize the food substances to release energy during respiration (1)

(d) To reduce overcrowding (1)

To reduce competition between individuals for light, water and minerals (1)

(Or other reasonable answers)

(e) When the pods containing the peas are mature, the pod wall will dry up and shrink. (1)

This causes the pods to split and this 'throws' the peas over a wide area. (1)

(f) Structure F (1)

(11 marks)

##

	Skill assessed	Marks
(a)	Analysing data	2
(b)	Analysing data	2
(c)	Understanding basic concepts	2
(d)	Understanding basic concepts	2
(e)	Understanding basic concepts	2
(f)	Analysing data	1

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